Computational Finance
Lecture 6

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Lecture 6 – Fourier methods for option pricing

- The Fourier family
- The COS method
- The COS method for option pricing

The Fourier family

- Continuous Fourier transform
- Discrete Fourier transform
  - FFT
- Fourier series expansion
  - Fourier-cosine series expansion
  - Fourier-sine series expansion

Part 1: The Fourier Family

Continuous Fourier transform

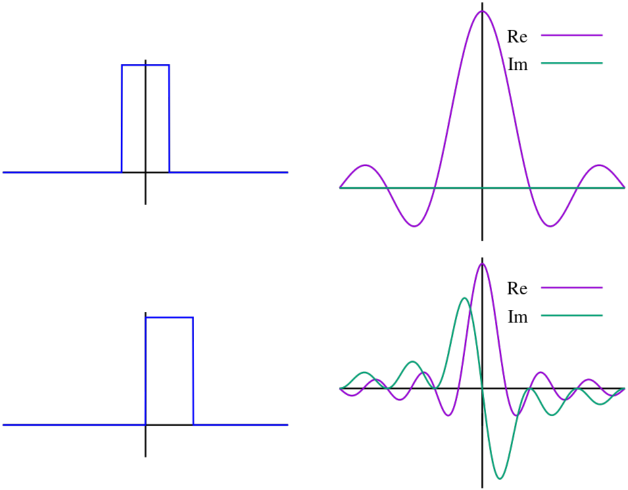
- Definition
  - There are several common conventions for defining the Fourier transform of an integrable function $f : \mathbb{R} \to \mathbb{C}$. One of them is:

$$\hat{f}(\omega) = \int_{-\infty}^{\infty} f(x) e^{i2\pi\omega x}\, dx, \qquad \forall \omega \in \mathbb{R}$$

  - The transform of function $f(x)$ at frequency ω is given by the complex number $\hat{f}(\omega)$. Evaluating this equation for all values of ω produces the frequency-domain function.
  - Under suitable conditions f can be represented as a recombination of complex exponentials of all possible frequencies:

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{f}(\omega) e^{-i2\pi\omega x}\, d\omega, \qquad \forall x \in \mathbb{R}$$

Example 1



Example 2 - Characteristic function (ch.f.)

- $\varphi_X(t)$, the ch.f. of a random variable X, is defined as

$$\varphi_X(t) = \mathbb{E}(e^{itX}) = \int_{-\infty}^{\infty} e^{itx} f_X(x)\, dx$$

  where i is the unit of the imaginary numbers.

- $F_X(x)$, the cumulative distribution function (CDF) of X, completely determines the characteristic function $\varphi_X(t)$; and vice versa.
- Particularly, if X admits a probability density function $f_X(x)$, then $\varphi_X(t)$ is its Fourier dual, in the sense that $f_X(x)$ and $\varphi_X(t)$ are the **Continuous Fourier transform** of each other.

Characteristic function (ch.f.): the Inversion formula

- The inversion from $\varphi_X(t)$ to $f_X(x)$ is given by the Fourier transform:

$$f_X(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-itx} \varphi_X(t)\, dt$$

- Before the invention of the COS method, the integral on the right-hand side was typically solved by numerical integration methods.

Ch.f. in option pricing models

- In many option pricing models, the ch.f. of the asset price or return is known analytically.
- A long list of models have ch.f. available analytically or semi-analytically:
  - Black–Scholes
  - Heston stochastic volatility model
  - Variance-Gamma process
  - Affine jump diffusion process in general

Discrete Fourier transform

- Definition
  - The discrete Fourier transform is an invertible, linear transformation $f : \mathbb{C}^N \to \mathbb{C}^N$, i.e. the discrete Fourier transform transforms a sequence of N complex numbers $\{\boldsymbol{x_k}\} := x_0, x_1, \ldots, x_{N-1}$ into another sequence of complex numbers, $\{\boldsymbol{X_k}\} := X_0, X_1, \ldots, X_{N-1}$.
  - One of the definition conventions is

$$X_k = \sum_{n=0}^{N-1} x_n e^{i\frac{2\pi}{N}kn}$$

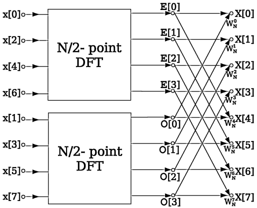
  - The inverse transform is defined as

$$x_n = \frac{1}{N} \sum_{k=0}^{N-1} X_k e^{-i\frac{2\pi}{N}kn}$$

  - Can also be obtained by discretizing the CFT.

FFT – Fast Fourier Transform

- Definition: a fast algorithm to compute the DFT.
  - An FFT rapidly computes such transformations by factorizing the DFT matrix into a product of sparse (mostly zero) factors. As a result, it manages to reduce the complexity of computing the DFT from $O(N^2)$ to $O(N \log N)$.
  - The difference in speed can be enormous, especially for long data sets where N may be in the thousands or millions.



Fourier series expansion

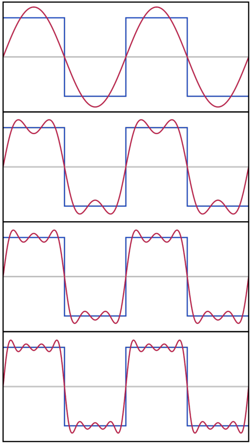
- Definition
  - A Fourier series is a sum that represents a periodic function as a sum of sine and cosine waves. The frequency of each wave in the sum, or harmonic, is an integer multiple of the periodic function's fundamental frequency.

$$f(x) = \frac{a_0}{2} + \sum_{n=0}^{\infty} a_n \cos(nx) + b_n \sin(nx)$$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos(nx)\,dx, \qquad b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin(nx)\,dx$$

Example

- https://en.wikipedia.org/wiki/Fourier_series#/media/File:Fourier_series_and_transform.gif



Fourier-cosine and Fourier-sine series

- Sine series
  - If $f(x)$ is an odd function with period $2L$, then the Fourier Half Range sine series of $f(x)$ is defined to be

$$f(x) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi}{L}x\right) \qquad b_n = \frac{2}{L}\int_0^L f(x)\sin\left(\frac{n\pi}{L}x\right)\,dx$$

- Cosine series
  - If $f(x)$ is an even function with period $2L$, then the Fourier Half Range sine series of $f(x)$ is defined to be

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi}{L}x\right) \qquad a_n = \frac{2}{L}\int_0^L f(x)\cos\left(\frac{n\pi}{L}x\right)\,dx$$

Expectation representation of option pricing

- The payoff function of a European Call option with maturity T and strike K is

$$V = \max(S_T - K, 0)$$

  - S_T is the asset price at maturity T.
  - V is the payoff to be delivered at time T.

- The option price at time $t = 0$ is the discounted risk neutral expectation of the payoff V.

$$P(x_0, T) = e^{-rT}\mathbb{E}[V(x_T, T) \mid x_0] = e^{-rT}\int V(x_T, T) f(x_T \mid x_0)\, dx_T$$

  - x_T is the state variable of the payoff function V, e.g., x_T could be S_T or $\log S_T$.
  - x_0 is the value of the state variable at time $t = 0$.
  - $f(x_T \mid x_0)$ is the conditional probability density of x_T given x_0 in the risk neutral probability measure.
  - r is the discount rate.

The first Fourier method in option pricing: Carr–Madan 1998 (1/3)

- Let $k = \ln K$ (the log of the strike), and $x_T = \ln S_T$ (the log of the asset price), the call price can be written as:

$$C_T(k) = e^{-rT} \int_k^{+\infty} (e^{x_T} - e^k) f(x_T \mid x_0)\, dx_T$$

- Because $\lim_{k \to -\infty} C_T(k) = S_0$ (due to the property of the risk neutral measure), $C_T(k)$ is not square integrable. We work with a modified call price $c_T(k)$ defined by:

$$c_T(k) = e^{\alpha k} C_T(k),$$

for some positive $\alpha > 0$.

The first Fourier method in option pricing: Carr–Madan 1998 (2/3)

- The Fourier transform of damped $c_T(k)$ is given by

$$\psi_T(v) = \int_{-\infty}^{+\infty} e^{\alpha k} e^{ivk} c_T(k)\, dk$$

- If $\psi_T(v)$ is known, we can obtain the call price numerically using the inverse Fourier transform:

$$C_T(k) = \frac{e^{-\alpha k}}{2\pi} \int_{-\infty}^{+\infty} e^{-ivk} \psi_T(v)\, dv$$

- The integral above can be numerically computed using FFT.

The first Fourier method in option pricing: Carr–Madan 1998 (3/3)

- A key step is to find out $\psi_T(v)$ in order to utilize FFT.
- It turns out that $\psi_T(v)$ can be expressed in terms of $\varphi_X(t)$ (the ch.f. of x_T):

$$\psi_T(v) = \int_{-\infty}^{+\infty} e^{ivk} \int_k^{+\infty} e^{\alpha k} e^{-rT} (e^{x_T} - e^k) f(x_T \mid x_0)\, dx_T\, dk = \frac{e^{-rT} \varphi_X(v - (\alpha+1)i)}{\alpha^2 + \alpha - v^2 + i(2\alpha+1)v}$$

- This is why the characteristic function methods are appealing.

Part 2: The COS Method

The COS method (1/2)

- The COS method (Fang–Oosterlee 2009), in essence, is a highly efficient semi-analytical algorithm to solve the expectation of a function of random variables.
- Unlike FFT, COS does not require any numerical integration and thus is much faster.
- For a smooth function supported on a finite interval $[a, b]$, the Fourier cosine expansion reads

$$f(x) = {\sum_{k=0}^{+\infty}}' A_k \cdot \cos\left(k\pi \frac{x-a}{b-a}\right),$$

where $\sum'$ indicates that the first term in the summation is weighted by one-half.

The COS method (2/2)

- The key insight of COS is that the coefficients of the cosine expansion A_k can be directly sampled from the Fourier transform of $f(x)$.

$$A_k = \frac{2}{b-a}\Re\left\{\varphi\left(\frac{k\pi}{b-a}\right)\cdot\exp\left(-i\frac{ka\pi}{b-a}\right)\right\}$$

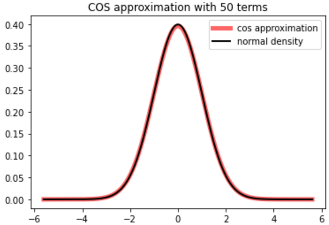
  - $\Re(\cdot)$ denotes the real part of a complex number.
  - $$\varphi(t) := \int_a^b e^{itx} f(x)\,dx$$

- For smooth probability densities $f(x)$ supported on $(-\infty,+\infty)$, we can truncate the distribution on a large enough interval $[a,b]$ and replace φ in the definition of A_k by the ch.f. of $f(x)$ as an accurate approximation.

- COS is proved to converge exponentially for regular densities.

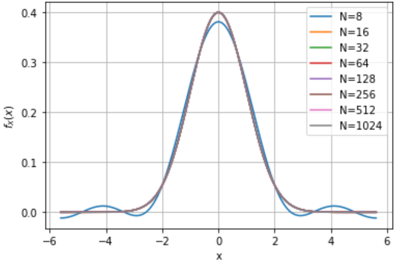
COS approximation of the standard Normal density function

- $$f(x) = \frac{A_0}{2} + \sum_{k=1}^{K} A_k \cdot \cos\left(k\pi \frac{x-a}{b-a}\right)$$
  - K: the number of COS terms.
  - a: the left endpoint of the truncation range, e.g., $N^{-1}(10^{-8})$.
  - b: the right endpoint of the truncation range, e.g., $-N^{-1}(10^{-8})$.
  - $$A_k = \frac{2}{b-a}\Re\left\{e^{-\frac{1}{2}\left(\frac{k\pi}{b-a}\right)^2} \cdot \exp\left(-i\frac{ka\pi}{b-a}\right)\right\}$$



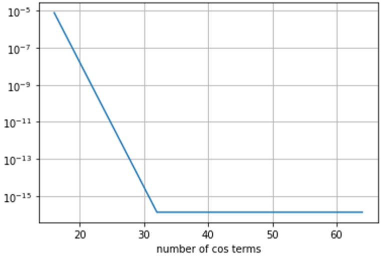
Convergence of COS approximation of the standard Normal density function (1/2)

- Fix the truncation range $a = -N^{-1}(10^{-8})$ and $b = -a$.
- As the number of COS terms increases, the accuracy improves.



Convergence of COS approximation of the standard Normal density function (2/2)

- At a fixed point of the density function, e.g., the 95% quantile, we observe the error convergence of COS is exponential.
- The error convergence stops at the level of the error caused by truncation range $[a, b]$.



Some examples of Ch.f.

Model	Characteristic function
BS	$\varphi(\xi,t)=\exp\left(i\xi\mu t-\frac{1}{2}\sigma^2\xi^2 t\right)$
NIG	$\varphi_{\mathrm{NIG}}(\xi,t;\alpha,\beta,\delta)=\exp\left[\delta t\left(\sqrt{\alpha^2-\beta^2}-\sqrt{\alpha^2-(\beta+i\xi)^2}\right)\right]$
Kou	$\varphi_{\mathrm{Kou}}(\xi,t;\lambda,p,\eta_1,\eta_2)=\exp\left[\lambda t\left(\frac{p\eta_1}{\eta_1-i\xi}+\frac{(1-p)\eta_2}{\eta_2+i\xi}-1\right)\right]$
Merton	$\varphi_{\mathrm{Merton}}(\xi,t;\lambda,\bar{\mu},\bar{\sigma})=\exp\left[\lambda t\left(\exp\left(i\bar{\mu}\xi-\frac{1}{2}\bar{\sigma}^2\xi^2\right)-1\right)\right]$
VG	$\varphi_{\mathrm{VG}}(\xi,t;\sigma,\nu,\theta)=\left(1-i\xi\theta\nu+\frac{1}{2}\sigma^2\nu\xi^2\right)^{-t/\nu}$
CGMY	$\varphi_{\mathrm{CGMY}}(\xi,t;C,G,M,Y)=\exp\left(Ct\Gamma(-Y)\left[(M-i\xi)^Y-M^Y+(G+i\xi)^Y-G^Y\right]\right)$

Source: Fang–Oosterlee 2009, *Pricing early-exercise and discrete barrier options by Fourier-cosine series expansions.*

Part 3: Apply the COS Method in Pricing European Options

COS: pricing European options (1/3)

- Recall that the price of a European option with maturity T and payoff $V(x_T, T)$ is

$$P(x_0, T) = e^{-rT}\mathbb{E}[V(x_T, T) \mid x_0] = e^{-rT}\int V(x_T, T)f(x_T \mid x_0)\, dx_T$$

- First truncate the density $f(x_T \mid x_0)$ with an interval $[a, b]$ such that $[a, b]$ covers a sufficiently large probability of x_T; and apply the Fourier-Cosine series expansion:

$$f(x_T \mid x_0) \approx {\sum_{k=0}^{+\infty}}' A_k(x_0) \cos\left(k\pi \frac{x_T - a}{b - a}\right)$$

COS: pricing European options (2/3)

- We could rewrite $P(x_0, T)$ as

$$P(x_0, T) \approx e^{-rT} \int_a^b V(x_T, T) \sum_{k=0}^{+\infty}{}' A_k(x_0) \cos\left(k\pi \frac{x_T - a}{b - a}\right) dx_T$$

- We interchange the summation and integration, and insert the definition

$$V_k := \frac{2}{b-a} \int_a^b V(x_T, T) \cos\left(k\pi \frac{x_T - a}{b - a}\right) dx_T$$

  - V_k has an analytical expression for the European call or put option.

COS: pricing European options (3/3)

- It results in transforming the product of $f(x_T \mid x_0)$ and $V(x_T, T)$ into that of their Fourier-cosine series coefficients:

$$P(x_0, T) \approx \frac{b-a}{2} e^{-rT} \sum_{k=0}^{+\infty}{}' A_k(x_0) V_k$$

- Next sample A_k directly from the ch.f. of x_T and truncate the series summation to obtain an approximation with finite number of terms:

$$P(x_0, T) \approx e^{-rT} \sum_{k=0}^{K-1}{}' \Re \left\{ \varphi_{X_T} \left(\frac{k\pi}{b-a}; x_0 \right) e^{-ik\pi \frac{a}{b-a}} \right\} \cdot V_k$$

Error convergence of COS

- Three sources of errors; each converges regularly.
    - Range truncation error of $[a, b]$.
    - Series truncation error of the Fourier cosine series.
- The COS method can exhibit exponential convergence when the density function is very smooth.

Exponential convergence

- Obviously COS is faster than Monte Carlo, which converges at rate $\sqrt{n}$.
- COS is also significantly faster than FFT methods.

TABLE 4

Error convergence and CPU times for the COS and Carr–Madan methods for calls under the Heston model with $T = 1$, with parameters as in (53); reference val. $= 5.785155450\ldots$.

COS			Carr–Madan		
N	Error	Time (msec.)	N	Error	Time (msec.)
64	−4.92e-03	0.61	256	−2.29e+06	4.70
96	−2.99e-04	0.78	512	2.31e+01	6.94
128	1.94e-05	0.94	1024	−2.61e-01	11.30
160	2.99e-06	1.11	2048	−2.14e-03	20.29
192	−3.17e-07	1.27	4096	3.76e-07	38.54

TABLE 5

Error convergence and CPU time for the COS and Carr–Madan methods for calls under the Heston model with $T = 10$, with parameters as in (53); reference val. $= 22.318945791\ldots$.

COS			Carr–Madan		
N	Error	Time (msec.)	N	Error	Time (msec.)
32	7.40e-03	0.46	128	−1.99e+06	3.64
64	−5.02e-05	0.62	256	1.36e+05	4.78
96	1.40e-07	0.81	512	3.27e+01	7.08
128	4.92e-10	0.98	1024	−2.61e-01	11.38
160	−1.85e-10	1.36	2048	−2.15e-03	20.93

Greeks

- Greeks can be easily derived from the COS price formula.
- Delta:
$$\frac{\partial P}{\partial S_0} \approx e^{-rT} \sum_{k=0}^{N-1}{}' \Re \left\{ \phi\left(\frac{k\pi}{b-a}; u_0\right) e^{ik\pi \frac{x-a}{b-a}} \cdot \frac{ik\pi}{b-a} \right\} \cdot \frac{V_k}{S_0}$$
- Vega:
$$\frac{\partial P}{\partial u_0} \approx e^{-rT} \sum_{k=0}^{N-1}{}' \Re \left\{ \frac{\partial \phi\left(\frac{k\pi}{b-a}; u_0\right)}{\partial u_0} e^{ik\pi \frac{x-a}{b-a}} \right\} V_k$$

More Applications in Computational Finance

- Calibration of option pricing models that have analytical or semi-analytical ch.f. (recent research in this direction includes, e.g., rough volatility models).
- Analytical calculation of Greeks.
- Calculation of credit portfolio loss distribution (Economic Capital, Default Risk Charge).
- Counterparty credit risk (CCR) and XVA.
- Training deep neural networks to accelerate pricing/calibration.
- Solving portfolio optimization problems (e.g., the problem in Brandt, Goyal, Santa-Clara and Storud, *Review of Financial Studies* 2005).

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